

Learning Plan Plugin

Complete Administrator Guide & Technical Architecture Reference

A complete end-to-end guide detailing pathway curriculum design, chapter and step creation, learner cohort enrollments, automated gamified badges engine, scoring synchronization, and real-time institutional reporting.

Manual Specifications

Plugin: local_learningplan | **Target Roles:** Site Administrators, Managers & Teachers

Status: Verified & Production Ready | **Engine Version:** 2.5 (Fully Gamified)

1. System Overview & Core Philosophy

The **Learning Plan Plugin** (local_learningplan) provides a modern, visual, and gamified curriculum delivery system within Moodle. Instead of navigating scattered courses and separate gradebooks, learners follow an adventure-style journey map composed of structured **Chapters** and **Step Nodes**.

Multi-Modal Step Types

Integrate whole courses, specific quizzes/assignments, file evidence, and external URLs into one cohesive curriculum.

Gamified Badges Engine

1-click auto-generation of active Moodle OpenBadges with baked PNG icons and instant milestone awarding.

Points (XP) & Stars

Custom XP point values and 1-3 star ratings awarded upon mastery with real-time class leaderboards.

Cascading Linear Unlocks

Enforce prerequisites: completing previous steps automatically unlocks subsequent milestones.

2. Learning Plans Management Hub (manage/plans.php)

The Management Hub serves as the administrative mission control where all curriculum pathways are organized, themed, and monitored.

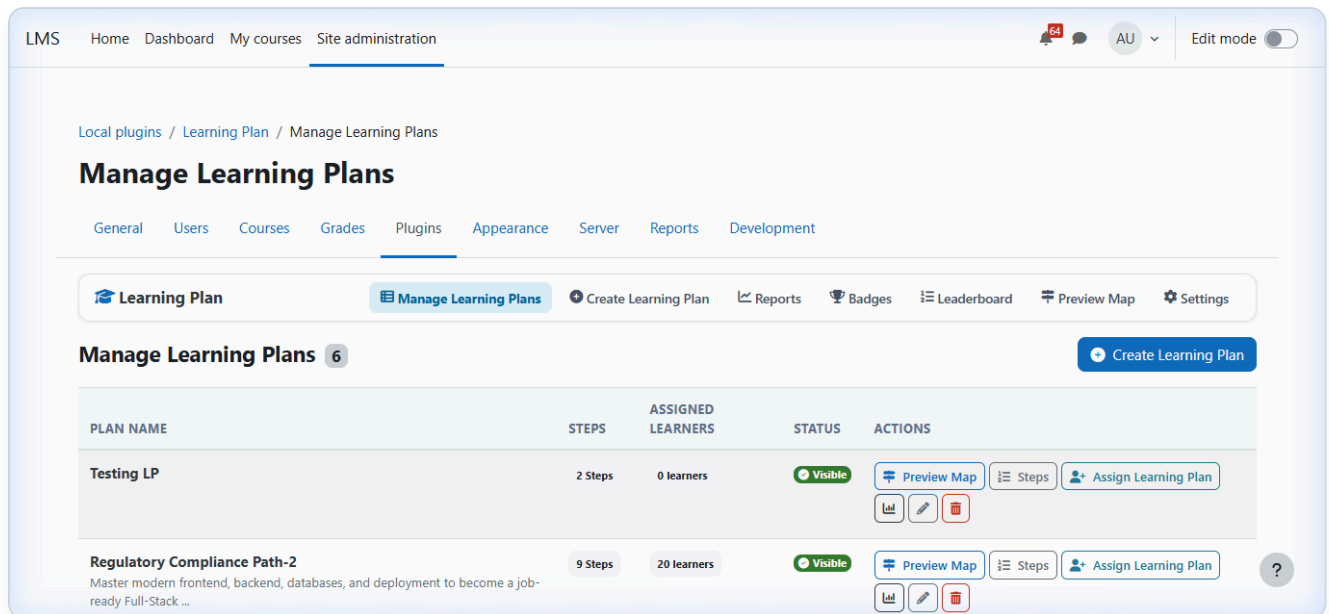


Figure 1: Learning Plans Administration Hub with plan cards, visual palettes, and quick actions.

Administrative Actions from the Hub:

- **Create Learning Plan:** Launch the pathway wizard to configure title, description, and visual theme (Ocean, Sunset, Forest, Candy).
- **Manage Steps:** Open the visual chapter and step builder for curriculum sequencing.
- **Assign Learners:** Enroll individual users or entire student cohorts.
- **Manage Badges:** Access auto-generation, custom badge creation, and retroactive award tools.
- **Reports & Leaderboard:** Inspect student progression tables, XP metrics, and download audit CSVs.
- **View Map:** Preview the full student interactive map interface in real time.

3. Chapter & Step Hierarchy Builder (manage/edit_steps.php)

Curriculum is structured in a two-tier hierarchy: **Chapters** (major modules/milestones) and **Steps** (individual learning actions).

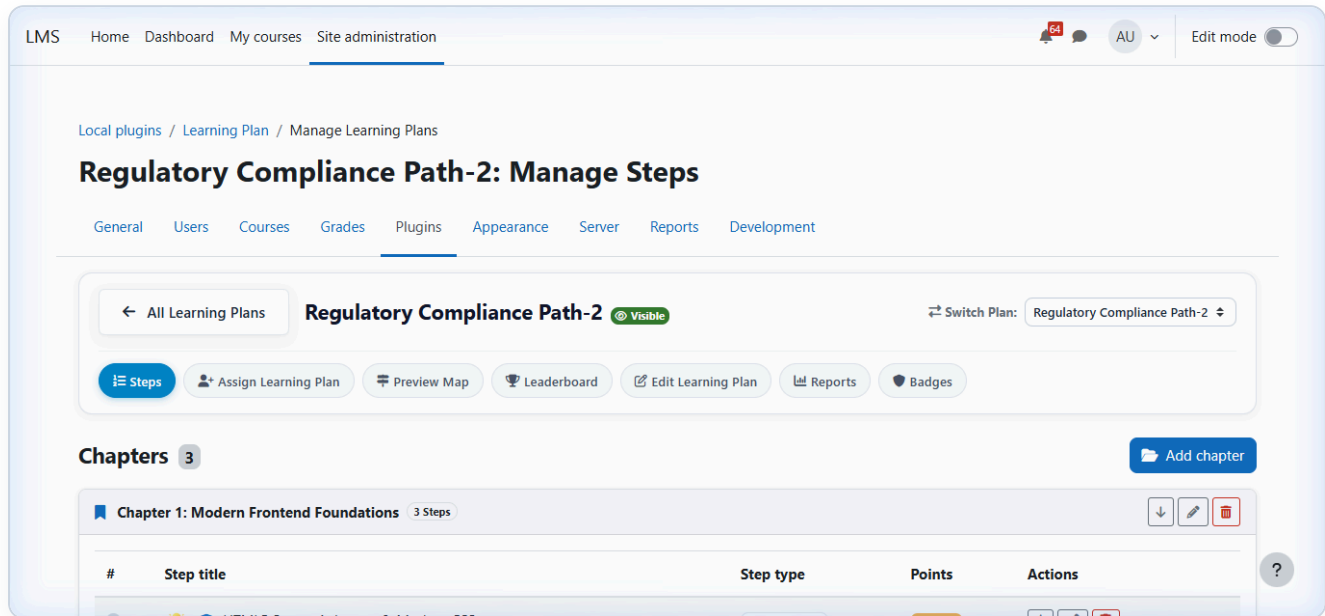


Figure 2: Chapter & Step Hierarchy Builder showing milestone organization, points badges, and sequencing.

Supported Step Types & Completion Criteria

Step Type	Completion & Unlock Logic	Configuration Parameters	Student Experience
 Course	Evaluates Moodle core course completion criteria via completion_info.	Moodle Course ID selection.	Redirects to course view; completes when course criteria met.
 Activity	Evaluates specific module (Quiz, Assignment, SCORM, Lesson) completion state.	Course Module ID (cmid) selector.	Direct link to activity; auto-completes on pass/grade.
 File Upload	Student uploads document/evidence (e.g. PDF/DOCX) in the step modal.	Title, prompt, XP points, star rating.	Uploads file inside map popup; marks complete instantly.
 External URL	External resource, documentation, or video link.	Destination URL, XP points, star rating.	Opens link in new tab; self-reported completion button.

4. Learner Assignment & Enrollment (manage/assign.php)

Enrolling students into a learning pathway initializes their adventure map, seeds step 1 to available, and unlocks previous achievements.

LMS
Home
Dashboard
My courses
Site administration
64
AU
Edit mode

Local plugins / Learning Plan / Manage Learning Plans

Regulatory Compliance Path-2: Assign Learning Plan

General
Users
Courses
Grades
Plugins
Appearance
Server
Reports
Development

← All Learning Plans
Regulatory Compliance Path-2
Visible
Switch Plan: Regulatory Compliance Path-2

Steps
Assign Learning Plan
Preview Map
Leaderboard
Edit Learning Plan
Reports
Badges

Assign Learning Plan
Assign to: By individual learner
Select learners

Assigned learners
20 records

Assign to	[[type]]	
Alex Rivera	By individual learner	Remove assignment
Ava Thomas	By individual learner	Remove assignment

Figure 3: Learner Assignment Hub supporting individual student enrollment and bulk cohort assignment.

⚙️ What Happens Upon Learner Enrollment

- Progress Table Seeding:** Step 1 of Chapter 1 is initialized to available; subsequent steps to locked.
- Scoring Sync Evaluation:** `scoring::sync_user()` inspects whether the student has previously completed any Moodle courses/modules linked in the pathway and unlocks them in order.
- Automatic Badge Issuance:** Any completed chapters immediately award their milestone badges.

5. Gamified Badges & Milestones Engine (manage/badges.php)

The Badges Engine seamlessly connects learning pathway achievements to Moodle's core OpenBadges system, issuing verifiable credentials.

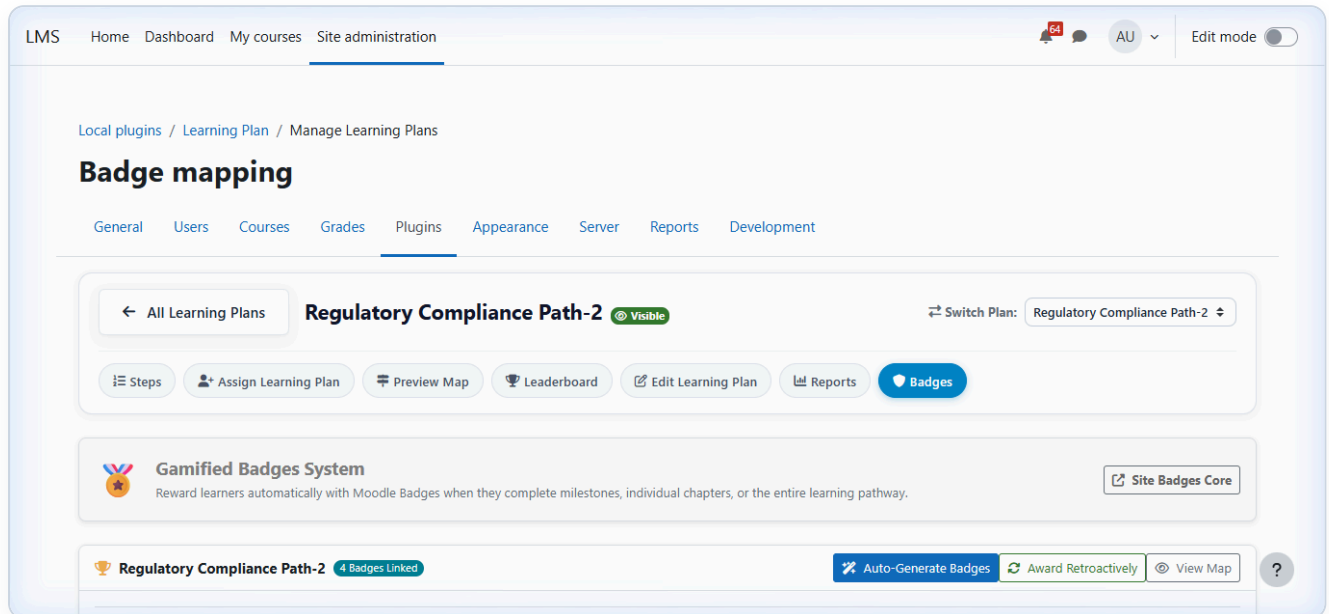


Figure 4: Badges Management Hub with 1-Click Auto-Generation, Retroactive Awarding, and side-by-side forms.

Key Administrative Badge Features:

- ⚡ **1-Click Auto-Generate Badges:** Inspects all plan chapters and automatically creates active Moodle site badges with custom artwork for every chapter milestone plus a Grand Champion badge for 100% completion.
- 🔄 **Award Retroactively:** Mass-syncs all enrolled students and issues badges to any learner who already completed criteria.
- 🔗 **Link an Existing Badge:** Attach any pre-created Moodle core badge to a chapter milestone or whole-plan completion.
- 🌟 **Quick Create Custom Badge:** Form featuring real-time live emblem artwork preview and instant auto-activation.

🔒 Idempotency & OpenBadges Baking Guarantee

The engine enforces zero duplicate badge issuance by checking `{badge_issued}` records prior to dispatch. All badges are issued with `$nobake = false` so PNG recipient metadata is baked and recipient notifications are dispatched.

6. Progress Tracking & Analytics Dashboard (manage/report.php)

The Institutional Progress Report provides teachers and administrators with real-time analytics across all assigned learners.

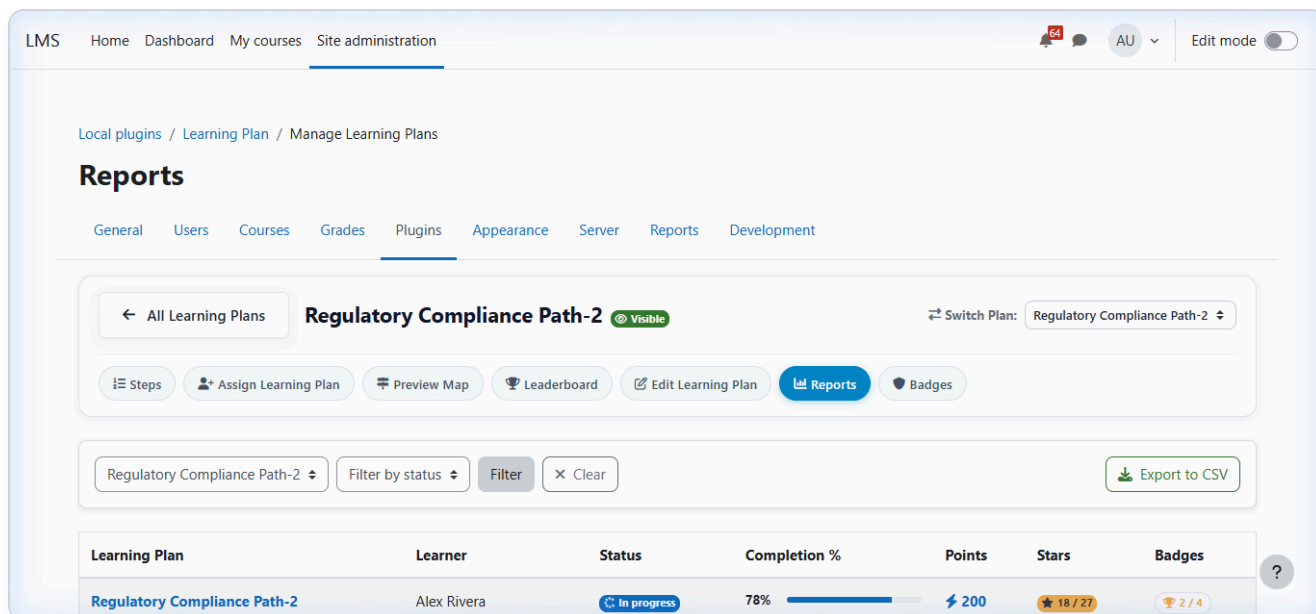


Figure 5: Institutional Progress Report with Badges metrics, progress bars, and CSV export.

7. Real-Time Leaderboard & Gamified Adventure Map

The learning plan motivates students through real-time rank podiums, XP scores, and an interactive adventure map interface.

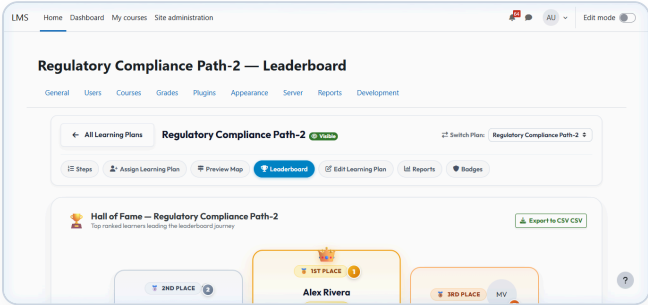


Figure 6: Real-time Gamified Leaderboard with Gold/Silver/Bronze ranks.

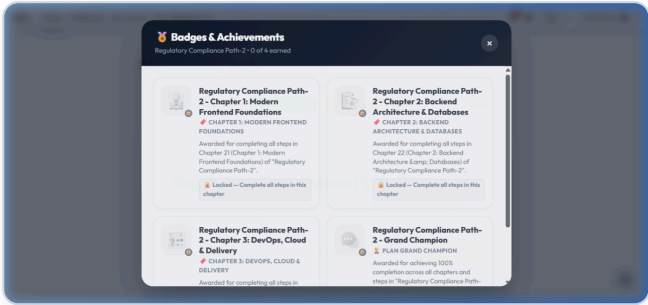


Figure 7: Learner Map & Badges & Achievements Glassmorphic Modal.

8. Technical Architecture & Database Schema

The plugin is designed for high performance, utilizing indexed relational tables and idempotent cascading triggers:

Table Name	Purpose & Responsibility	Key Columns & Constraints
mdl_local_learningplan_plan	Core pathway definition records.	id, name, description, coverimage, timecreated
mdl_local_learningplan_chapter	Chapter milestone containers.	id, planid, title, sortorder, description
mdl_local_learningplan_step	Individual step node configuration.	id, chapterid, title, steptype, courseid, cmid, points, maxstars, icon
mdl_local_learningplan_user	Learner-to-Plan enrollment records.	id, planid, userid, timeenrolled
mdl_local_learningplan_progress	Step completion status & stars earned.	id, stepid, userid, status, starsearned, timecompleted
mdl_local_learningplan_badge	Milestone to badge mapping definitions.	id, planid, chapterid, badgeid, criteria
mdl_local_learningplan_score_log	Audit trail of XP points & star transactions.	id, planid, stepid, userid, points, starsearned, timecreated

✓ **End of Administration Guide**

For technical assistance or custom badge extensions, contact your Moodle system administrator or reference the official plugin documentation in `local/learningplan/README.md`.